Marco Del Fava

Research Software & Neural Data

M.S. Biomedical Engineering student | UC Davis | Human intracranial electrophysiology

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Engineering Summary

Biomedical Engineering M.S. student and neuroscience researcher at UC Davis. I build Python acquisition and data-integrity workflows and use coding agents with profiling, debugging, and regression tests.

Technical Skills

Scientific Programming: Python, NumPy, Pandas, scikit-learn, pytest, Git, Linux, Docker; MATLAB

Neural Data & Acquisition: Neuralynx, Intan RHD, DarkHorse MED, Neo, SpikeInterface, Kilosort4, MountainSort5

Engineering Methods: Scientific data parsing, metadata validation and repair, waveform analysis, experimental synchronization, profiling, regression tests

Research Experience

Graduate Student, UC Davis Health — Z-Lab (Dr. Jie Zheng) — Jul 2025 - Present

- Work with human intracranial electrophysiology recordings, developing Python workflows to prepare, process, and analyze single-neuron data.

Selected Engineering Projects

Neural Data Formats and Integrity

- Adapted a Neo-backed MED reader to skip corrupt parity channels whose metadata caused standard readers to abort, allowing processing without modifying source recordings.
- Evaluated signal fidelity across 20 million samples on one channel (32 kHz; 625 s), comparing original CSC and compressed MED recordings. Measured a maximum amplitude difference of 0.5 µV, with exact agreement after whole-microvolt source rounding.

Neural Acquisition and Experimental Synchronization

- Built Python workflows for processing Intan recordings through SpikeInterface, including probe mapping, channel checks, spike sorting, waveform extraction, and quality metrics.
- Developed synchronization software to align PsychoPy behavioral events with RHX recordings, using matched event labels and affine clock fitting to characterize drift between recording systems.

Debugging Agent-Generated Software

- Investigated a performance regression in Hosaka, a command-line software platform, where agent-generated schema validation made listing seven records take approximately 70 seconds.
- Used execution timing logs to identify repeated validation as the bottleneck, then introduced content-based caching and regression tests to preserve checks for changed or invalid schemas.

Additional Professional Experience

IT Administrator, First Step Communities — Jul 2021 - Present

- Maintain network and IT infrastructure across multiple shelter sites, including routing/VLAN segmentation, Microsoft 365 administration, and troubleshooting.

Analyst, Eurofins Test America — Oct 2019 - Jun 2021

- Prepared soil and water samples using solvent extraction in a high-throughput environmental testing laboratory, following established quality assurance procedures.

Education

M.S. Biomedical Engineering, University of California, Davis — 2026–Present

M.S. Psychology, Arizona State University — 2023

B.S. Neuroscience, Physiology, and Behavior, University of California, Davis — 2017